UNIFIED MECHANICS

Derivation Programme

From Electromagnetism to Complete Physical Closure

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Abstract

This dossier follows every node and connection in the supplied unification map from inside the Unified Mechanics theory. It shows how the electromagnetic, electroweak, strong, Standard-Model, grand-unified, gravitational, cosmological, quantum, and observational sectors inhabit one commuting construction. Unconditional mathematics, conditional physical realization, interpretation-dependent branches, and localized frontier calculations remain separately graded. A presently non-unique interpretation remains under determination unless an explicit incompatibility or empirical exclusion is established under declared premises.

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CHAPTER 1

The derivation contract and the supplied map

1.1. What has been supplied

The input is Unified Mechanics as a theory of everything, together with its mathematical framework, internal constructions, and the unification map reproduced in figure 1.1. The boxes name the physical domains carried by the theory. A line is treated as a *derivation trace*: it must identify the precise map between mathematical structures, controlled limits, or symmetry sectors that makes the connection scientific rather than merely diagrammatic.

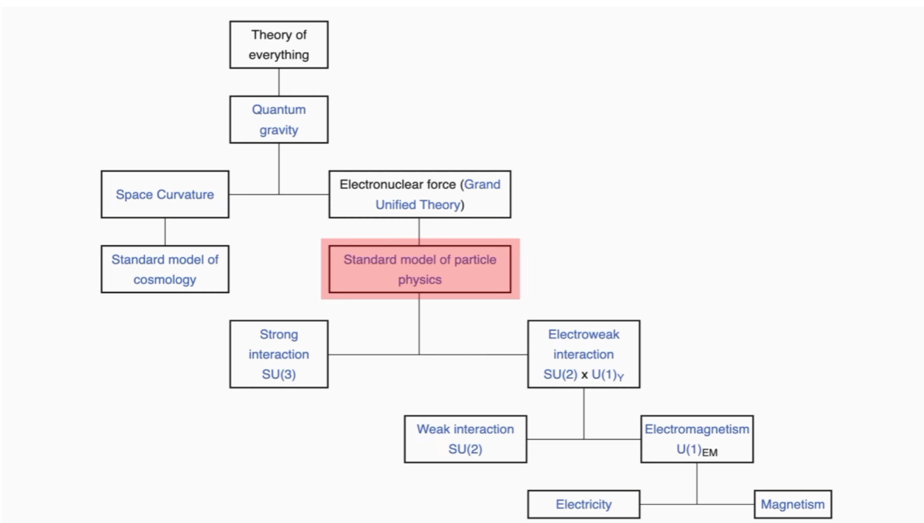

FIGURE 1.1. The supplied unification map. The diagram contains twelve labelled boxes and eleven intended relations. Line direction is inferred from vertical placement because the source contains no arrowheads.

INTERPRETATION WITHOUT SEMANTIC DEMOTION

The theory's status is not averaged from the status of every sentence inside it. Internal structures are calculated at their registered grade, and physical identifications are *stated with their premises. When several realizations survive, at most three are carried forward and the observable capable of separating them is stated. "Open" means not yet selected by the present premises; it does not mean that the surrounding theory is absent.*

1.2. The twelve nodes

Ordered from the most encompassing target to the most directly observable decomposition, the nodes are:

1. theory of everything;
2. quantum gravity;
3. spacetime curvature;
4. electronuclear force or grand unified theory;
5. the standard model of cosmology;
6. the Standard Model of particle physics;
7. the strong interaction, labelled $SU(3)$;
8. the electroweak interaction, labelled $SU(2) \times U(1)_Y$;
9. the weak interaction, labelled $SU(2)$;
10. electromagnetism, labelled $U(1)_{\mathrm{EM}}$;
11. electricity;
12. magnetism.

The labels are preserved, but the derivation uses the more precise conventional notation $SU(3)_c \times SU(2)_L \times U(1)_Y$. The subscripts matter: colour, left-chiral weak isospin, and hypercharge are physically distinct representation data.

1.3. Four meanings of a connecting line

A single undifferentiated hierarchy would obscure the mathematics. Each line is assigned one of four typed meanings.

Decomposition.
A structure has mutually coupled or direct-product sectors. The Standard Model gauge algebra decomposes as

$$\mathfrak{g}_{\mathrm{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y.$$

Symmetry reduction.
A vacuum or order parameter preserves a subgroup of a larger group. Electroweak symmetry reduction preserves $U(1)_{\mathrm{EM}}$, and a grand-unified model must preserve the Standard Model group at lower energy.

Controlled limit.
: A domain emerges after a scale, state, or symmetry restriction. Homogeneous and isotropic cosmology is a symmetry-reduced sector of gravitational dynamics; classical curvature must be a limit of any quantum-gravitational realization.

Closure node.
: Two or more domains inhabit one consistent construction even when a local realization or numerical selector remains non-unique. Quantum gravity and the theory-of-everything node have this type.

INTERNAL RESULT 1.1 — THE DIAGRAM IS A TYPED DEPENDENCY GRAPH

After relation typing, the apparent hierarchy becomes a directed acyclic derivation graph. Its lower branches admit direct derivations from established gauge and geometric structures. Its upper branches impose closure requirements on the lower ones. The theory-of-everything status is carried by the connected construction and its commuting limits, not by the label of any single box.

1.4. What counts as a derivation

For a proposed realization map

$$\mathscr{R} : \mathscr{F}_{\mathrm{UM}} \longrightarrow \mathscr{P}, \tag{1.1}$$

from framework objects $\mathscr{F}_{\mathrm{UM}}$ to physical objects $\mathscr{P}$, a derivation entry is the tuple

$$\mathfrak{D} = (I, A, C, R, O, S), \tag{1.2}$$

where I lists inherited premises, A additional assumptions, C the calculation, R the surviving realization maps, O observable discriminants, and S the scientific status.

Three different achievements must not be conflated:

$$\begin{aligned} \text{reconstruction} &: \quad \mathscr{R}(x) = p_{\text{known}}, \\ \text{retrodiction} &: \quad p_{\text{known}} \text{ influenced the selection of } \mathscr{R}, \\ \text{prediction} &: \quad \mathscr{R} \text{ and its parameters were fixed before } p \text{ was measured.} \end{aligned} \tag{1.3}$$

All three can be useful. Only the third supplies genuinely out-of-sample evidence.

1.5. Status language

Table 1.1. Statuses used throughout the derivation programme.

Status	Meaning
Derived	The conclusion follows from displayed premises.
Conditionally derived	The conclusion follows after a named physical-identification premise is added.
Interpretation-dependent	More than one internally coherent realization remains.
Open under present premises	A unique continuation has not yet been established; no contradiction is inferred.
Contradicted	An explicit inconsistency or quantitatively excluded consequence has been demonstrated.
Superseded	A later formulation replaces the earlier one while preserving its provenance.

1.6. Closure obligations omitted by the picture

The diagram names the principal sectors, but complete physical closure also requires cross-edges:

1. matter coupled to curvature through a defined stress–energy tensor;
2. quantum fields on nontrivial backgrounds and their renormalized backreaction;
3. thermal and cosmological history connecting particle dynamics to observations;
4. a global gauge-group choice, not only a Lie algebra;
5. state preparation, observables, probabilities, and records;
6. dimensional constants and a scale-setting mechanism;
7. a likelihood or other empirical comparison map.

These are not objections external to the framework. They are the mathematical interfaces through which the named sectors describe one another internally.

CHAPTER 2

Electromagnetism, electricity, and magnetism

This chapter closes the lowest branch of the diagram. Its conclusion is stronger than a verbal unification: electricity and magnetism are observer-dependent components of one antisymmetric spacetime field.

2.1. Minimal geometric data

Let M be an oriented four-dimensional Lorentzian manifold with metric signature $(-+++)$. Let $A \in \Omega^1(M)$ be a real one-form connection for a $U(1)$ gauge bundle. Its curvature is

$$F = \mathrm{d}A, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{2.1}$$

Because $\mathrm{d}^2 = 0$,

$$\mathrm{d}F = 0 \quad \Longleftrightarrow \quad \nabla_{[\lambda} F_{\mu\nu]} = 0. \tag{2.2}$$

This identity requires no field equation.

The gauge transformation

$$A \longmapsto A + \mathrm{d}\chi \tag{2.3}$$

leaves F invariant. With a conserved current one-form J, the action is

$$S_{\mathrm{EM}}[A] = -\frac{1}{2\mu_0}\int_M F \wedge \star F + \int_M A \wedge \star J. \tag{2.4}$$

This is the differential-form version of Maxwell electrodynamics (Maxwell, 1873).

2.2. Variation of the action

For a compactly supported variation $A \mapsto A + \varepsilon\,\delta A$, one has $\delta F = \mathrm{d}\delta A$. Therefore

$$
\begin{aligned}
\delta S_{\mathrm{EM}} &= -\frac{1}{\mu_0}\int_M \mathrm{d}(\delta A)\wedge\star F + \int_M \delta A\wedge\star J \\
&= -\frac{1}{\mu_0}\int_{\partial M}\delta A\wedge\star F + \int_M \delta A\wedge\left(\frac{1}{\mu_0}\mathrm{d}\star F + \star J\right).
\end{aligned}
\tag{2.5}
$$

With the boundary variation fixed, stationarity gives

$$
\mathrm{d}\star F = -\mu_0 \star J. \tag{2.6}
$$

The overall sign depends on signature and current conventions; the component convention adopted below is

$$
\nabla_\mu F^{\mu\nu} = \mu_0 J^\nu. \tag{2.7}
$$

Taking a divergence and using antisymmetry gives

$$
\nabla_\nu J^\nu = \frac{1}{\mu_0}\nabla_\nu\nabla_\mu F^{\mu\nu} = 0. \tag{2.8}
$$

Equivalently, invariance of the source term under equation (2.3) requires current conservation after integration by parts.

Derived result 2.1 — Maxwell closure

Given a Lorentzian metric, a $U(1)$ connection, the quadratic local action equation (2.4), and a conserved source, variation produces the inhomogeneous Maxwell equation. The homogeneous equation follows identically from $F = \mathrm{d}A$. Gauge invariance and charge conservation are mutually consistent parts of the same construction.

2.3. Observer-dependent split into electricity and magnetism

Let u^μ be a future-directed unit timelike vector, $u_\mu u^\mu = -1$, representing an observer congruence. Define

$$
E_\mu = F_{\mu\nu}u^\nu, \qquad B_\mu = (\star F)_{\mu\nu}u^\nu = \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}u^\nu F^{\rho\sigma}. \tag{2.9}
$$

Antisymmetry implies

$$
E_\mu u^\mu = B_\mu u^\mu = 0, \tag{2.10}
$$

so both fields are spatial relative to u. Conversely, the pair (E, B) reconstructs the six independent components of F:

$$F_{\mu\nu} = 2u_{[\mu}E_{\nu]} + \epsilon_{\mu\nu\rho\sigma}u^{\rho}B^{\sigma}, \tag{2.11}$$

with the sign of the second term fixed by the orientation and dual convention.

In a local inertial frame with $u^{\mu} = (1, 0, 0, 0)$, the covariant equations become

$$\begin{aligned} \nabla\cdot\mathbf{E} &= \rho/\epsilon_0, & \nabla\times\mathbf{B} - \frac{1}{c^2}\partial_t\mathbf{E} &= \mu_0\mathbf{J}, \\ \nabla\cdot\mathbf{B} &= 0, & \nabla\times\mathbf{E} + \partial_t\mathbf{B} &= 0. \end{aligned} \tag{2.12}$$

For an observer boosted by velocity $\mathbf{v}$, the parallel components remain unchanged while the transverse components mix:

$$\begin{aligned} \mathbf{E}'_{\parallel} &= \mathbf{E}_{\parallel}, & \mathbf{B}'_{\parallel} &= \mathbf{B}_{\parallel}, \\ \mathbf{E}'_{\perp} &= \gamma(\mathbf{E} + \mathbf{v}\times\mathbf{B})_{\perp}, & \mathbf{B}'_{\perp} &= \gamma\left(\mathbf{B} - \frac{1}{c^2}\mathbf{v}\times\mathbf{E}\right)_{\perp}. \end{aligned} \tag{2.13}$$

DERIVED RESULT 2.2 — THE TWO LOWEST BOXES ARE ONE FIELD

Electricity and magnetism are not separate fundamental interactions in this construction. They are the u-dependent decomposition of F. Changing u changes the split but not the spacetime tensor. The bottom line in the supplied diagram is therefore exactly realized as observer-dependent decomposition.

2.4. Invariant content and admissible interpretations

The two Lorentz invariants

$$I_1 = \frac{1}{2}F_{\mu\nu}F^{\mu\nu} = B^2 - \frac{E^2}{c^2}, \qquad I_2 = \frac{1}{4}F_{\mu\nu}(\star F)^{\mu\nu} = -\frac{\mathbf{E}\cdot\mathbf{B}}{c} \tag{2.14}$$

classify local field types independently of the observer. The stress–energy tensor is

$$T^{\mathrm{EM}}_{\mu\nu} = \frac{1}{\mu_0}\left(F_{\mu\alpha}F_{\nu}{}^{\alpha} - \frac{1}{4}g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right). \tag{2.15}$$

This tensor supplies the direct matter–curvature cross-edge used later.

The Unified Mechanics vocabulary permits three progressively stronger readings:

1. **Geometric reading:** the carrier connection has an Abelian component identified with A, so its curvature supplies F.
2. **Operational reading:** F is the channel-independent field, while (E, B) are observer-resolved readouts determined by u.

3. **Constitutive reading:** retention or boundary structure modifies the vacuum constitutive relation $G = \mu_0^{-1} \star F$, producing measurable polarization, dispersion, or boundary effects.

The first two are mutually compatible. The third adds physical content and must specify the map

$$G = \mathscr{C}[F;\ g, \mathscr{B}, \mathscr{R}] \tag{2.16}$$

for boundary data $\mathscr{B}$ and retained structure $\mathscr{R}$. In vacuum Maxwell theory, $\mathscr{C} = \mu_0^{-1}\star$.

CONDITIONAL RESULT 2.3 — FRAMEWORK REALIZATION OF ELECTROMAGNETISM

If the carrier contains a surviving compact Abelian connection whose matter representation has generator Q, then its curvature and minimal coupling reproduce the kinematic structure of electromagnetism. A distinct physical prediction arises only if the framework also fixes charge normalization, coupling evolution, global bundle sectors, or a non-Maxwell constitutive map.

2.5. Residual obligations for this branch

The geometric electric–magnetic relation is closed. The remaining framework-specific tasks occur higher in the diagram:

1. derive the unbroken generator Q from electroweak symmetry reduction;
2. derive the photon connection as the massless neutral combination;
3. determine electric-charge normalization from the matter embedding;
4. specify whether equation (2.16) differs from Maxwell vacuum electrodynamics;
5. if it differs, calculate a quantitative discriminant before comparison with the selected experiment.

Electroweak reduction to weak and electromagnetic sectors

The next line in the diagram is not an ordinary direct-product decomposition. The gauge symmetry $SU(2)_L \times U(1)_Y$ is realized in a vacuum that preserves one generator. Three gauge directions acquire masses and one remains massless. This is the electroweak mechanism developed through the work of Glashow, Weinberg, Salam, Englert, Brout, Higgs, Guralnik, Hagen, and Kibble (Glashow, 1961; Weinberg, 1967; Englert and Brout, 1964; Higgs, 1964; Guralnik et al., 1964).

3.1. Gauge and scalar data

Let

$$G_{\mathrm{EW}} = SU(2)_L \times U(1)_Y \tag{3.1}$$

with gauge fields W_μ^a and B_μ, couplings g and g', generators $T^a = \tau^a/2$, and hypercharge Y. For a field in a specified representation,

$$D_\mu = \partial_\mu - igT^a W_\mu^a - ig'YB_\mu. \tag{3.2}$$

The field strengths are

$$\begin{aligned} W_{\mu\nu}^a &= \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + g\epsilon^{abc} W_\mu^b W_\nu^c, \\ B_{\mu\nu} &= \partial_\mu B_\nu - \partial_\nu B_\mu. \end{aligned} \tag{3.3}$$

Take a complex scalar doublet with $Y_\Phi = 1/2$,

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \qquad V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda(\Phi^\dagger \Phi)^2, \qquad \mu^2, \lambda > 0. \tag{3.4}$$

The minima obey

$$\Phi^\dagger \Phi = \frac{\mu^2}{2\lambda} \equiv \frac{v^2}{2}. \tag{3.5}$$

After a gauge choice, a representative vacuum is

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}. \tag{3.6}$$

3.2. The unbroken generator

A generator $Q = aT^3 + bY$ is unbroken if it annihilates the vacuum. Since the lower component has $T^3 = -1/2$ and $Y = 1/2$,

$$Q\langle\Phi\rangle = \frac{-a+b}{2}\langle\Phi\rangle = 0 \quad\Longrightarrow\quad a = b. \tag{3.7}$$

Choosing the conventional normalization gives

$$Q = T^3 + Y. \tag{3.8}$$

Derived result 3.1 — uniqueness of the surviving local direction

For a single nonzero $SU(2)$ doublet vacuum of hypercharge $1/2$, the one-dimensional Lie-algebra subspace that annihilates the vacuum is generated by $Q = T^3 + Y$. Thus $U(1)_{\mathrm{EM}}$ is the stabilizer of the chosen vacuum, up to global-group and normalization data treated later.

3.3. Gauge-boson mass matrix

In unitary gauge write

$$\Phi(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}. \tag{3.9}$$

Substitution into $(D_\mu\Phi)^\dagger(D^\mu\Phi)$ gives the vector-boson mass terms

$$\mathscr{L}_{\mathrm{mass}} = \frac{g^2v^2}{4}W^+_\mu W^{-\mu} + \frac{v^2}{8}(gW^3_\mu - g'B_\mu)^2, \tag{3.10}$$

where

$$W^\pm_\mu = \frac{W^1_\mu \mp iW^2_\mu}{\sqrt{2}}. \tag{3.11}$$

The neutral mass-squared matrix, defined by

$$\mathscr{L}_{\mathrm{neutral,mass}} = \frac{1}{2}\begin{pmatrix} W^3_\mu & B_\mu \end{pmatrix} M_0^2 \begin{pmatrix} W^{3\mu} \\ B^\mu \end{pmatrix}, \tag{3.12}$$

is

$$M_0^2 = \frac{v^2}{4}\begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix}. \tag{3.13}$$

Its determinant vanishes and its eigenvalues are

$$m_A^2 = 0, \qquad m_Z^2 = \frac{v^2}{4}(g^2 + g'^2). \tag{3.14}$$

Define the weak mixing angle by

$$\tan\theta_W = \frac{g'}{g}, \qquad s_W = \sin\theta_W, \qquad c_W = \cos\theta_W. \tag{3.15}$$

The normalized eigenvectors are

$$\begin{aligned} A_\mu &= s_W W^3_\mu + c_W B_\mu, \\ Z_\mu &= c_W W^3_\mu - s_W B_\mu. \end{aligned} \tag{3.16}$$

Together with the charged sector,

$$m_W = \frac{gv}{2}, \qquad m_Z = \frac{v}{2}\sqrt{g^2 + g'^2}, \qquad \frac{m_W}{m_Z} = c_W \tag{3.17}$$

at tree level.

3.4. Photon coupling and electric charge

The neutral part of the covariant derivative becomes

$$\begin{aligned} gT^3 W^3_\mu + g'YB_\mu = \big(gs_W T^3 + g'c_W Y\big)A_\mu \\ + \big(gc_W T^3 - g's_W Y\big)Z_\mu. \end{aligned} \tag{3.18}$$

Since

$$e = gs_W = g'c_W, \tag{3.19}$$

the photon term is $e(T^3 + Y)A_\mu = eQA_\mu$. The Z coupling can be written

$$\frac{g}{c_W}\left(T^3 - s_W^2 Q\right) Z_\mu. \tag{3.20}$$

DERIVED RESULT 3.2 — ELECTROWEAK ORIGIN OF ELECTROMAGNETISM

Given equations (3.1), (3.4) and (3.6), diagonalization forces one neutral massless connection A_μ, three massive vector directions $W^\pm, Z$, the unbroken generator $Q = T^3 + Y$, and the coupling identity $e = gs_W = g'c_W$. This explicitly derives the electroweak-to-electromagnetic line in the source diagram.

3.5. Weak currents and the Fermi limit

The charged-current interaction has the form

$$\mathscr{L}_{\mathrm{CC}} = -\frac{g}{\sqrt{2}}\left(J^+_\mu W^{+\mu} + J^-_\mu W^{-\mu}\right). \tag{3.21}$$

At momentum transfer $q^2 \ll m_W^2$, the W propagator reduces to a local contact interaction. Matching to the conventional four-fermion form gives

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8m_W^2} = \frac{1}{2v^2}, \qquad G_F = \frac{1}{\sqrt{2}v^2}. \tag{3.22}$$

Thus the low-energy weak scale is the inverse square of the vacuum scale v, modulo radiative corrections when related to measured observables.

The label "weak interaction $SU(2)$" in the image has two coherent readings:

1. the pre-reduction $SU(2)_L$ gauge factor generated by T^a;
2. the broken-phase weak interactions mediated by $W^\pm$ and Z.

These readings are related but not identical. After reduction, Z contains both W^3 and hypercharge components, so the observed neutral weak current is not generated by an isolated exact $SU(2)$ factor.

3.6. What the framework must derive rather than import

The calculation above is conditional on the electroweak group, scalar representation, potential, and vacuum. The internal framework can deepen it at four points:

1. derive why the admissible local algebra contains $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$;
2. derive why the order parameter transforms as a doublet with $Y = 1/2$, or derive an observationally equivalent alternative;
3. derive v, λ, g, g' or relations among them from carrier invariants and renormalization conditions;
4. derive the chiral matter representations on which T^3 and Y act.

Three interpretations are retained while this mapping is developed:

Fundamental scalar interpretation.
: The carrier contains a primitive scalar order parameter whose lowest doublet is Φ.

Composite order-parameter interpretation.
: Φ is an effective bilinear or collective coordinate of deeper carrier degrees of freedom.

Geometric connection interpretation.
: Symmetry reduction is a restriction of a larger connection or generalized curvature, with Φ arising from an internal component or coset direction.

Conditional result 3.3 — three admissible realization classes

All three interpretations reproduce the tree-level reduction if their low-energy effective action contains equation (3.4). They separate through additional scalar states, form factors and coupling deviations, or geometric relations among the scalar and gauge sectors. Later carrier and unification constraints must reduce this list rather than selecting one by terminology alone.

3.7. Residual obligations

This branch has a complete conditional derivation but not yet a unique internal realization. The remaining obligations are to derive the electroweak input data, incorporate fermion masses and mixing, calculate renormalization-group evolution, and identify the global gauge group. Those questions are carried into the Standard Model and grand-unification chapters.

CHAPTER 4

The strong sector and the complete Standard Model

4.1. The $SU(3)_c$ connection

Let T^a, $a = 1, \dots, 8$, be Hermitian generators of $\mathfrak{su}(3)$ in the fundamental representation, normalized by

$$\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}, \qquad [T^a, T^b] = i f^{abc} T^c. \tag{4.1}$$

The colour connection and curvature are

$$\begin{aligned} G_\mu &= G_\mu^a T^a, \\ G_{\mu\nu} &= \partial_\mu G_\nu - \partial_\nu G_\mu - i g_s [G_\mu, G_\nu], \\ G_{\mu\nu}^a &= \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc} G_\mu^b G_\nu^c. \end{aligned} \tag{4.2}$$

The commutator term distinguishes the non-Abelian Yang–Mills structure from the Abelian field of Chapter 2 (Yang and Mills, 1954).

For quarks q_f in the fundamental representation,

$$\mathscr{L}_{\mathrm{QCD}} = -\frac{1}{4} G_{\mu\nu}^a G^{a\mu\nu} + \sum_f \bar{q}_f (i\gamma^\mu D_\mu - m_f) q_f + \frac{\theta_s g_s^2}{32\pi^2} G_{\mu\nu}^a \widetilde{G}^{a\mu\nu}, \tag{4.3}$$

where $D_\mu = \partial_\mu - i g_s G_\mu^a T^a$. The final term is locally a total derivative but can have physical consequences because the gauge configuration space has nontrivial topological sectors.

4.2. Running and the two dynamical regimes

For n_f active Dirac flavours, the one-loop beta function is

$$\mu \frac{\mathrm{d} g_s}{\mathrm{d}\mu} = -\frac{g_s^3}{(4\pi)^2} b_0 + O(g_s^5), \qquad b_0 = \frac{11}{3} C_A - \frac{4}{3} T_F n_f = 11 - \frac{2}{3} n_f, \tag{4.4}$$

using $C_A = 3$ and $T_F = 1/2$ for $SU(3)$. Hence $b_0 > 0$ for $n_f \leq 16$, and the coupling decreases at high scale (Gross and Wilczek, 1973; Politzer, 1973). Integration gives

$$\alpha_s(Q^2) \equiv \frac{g_s^2(Q^2)}{4\pi} = \frac{4\pi}{b_0 \ln\left(Q^2/\Lambda_{\mathrm{QCD}}^2\right)} \tag{4.5}$$

within its perturbative regime.

DERIVED RESULT 4.1 — ASYMPTOTICALLY FREE STRONG BRANCH

The gauge-boson self-interaction fixes the positive $11C_A/3$ contribution to b_0. With the observed number of quark flavours, it dominates the fermion screening term. The strong sector therefore possesses a controlled weak-coupling ultraviolet regime and a dynamically generated infrared scale.

The infrared regime is nonperturbative. Confinement, the mass gap, chiral symmetry breaking, and the hadron spectrum are calculable through lattice and other nonperturbative formulations, but they do not follow from truncating equation (4.5). Within the present derivation programme they are *open under analytic continuation from the displayed perturbative premises*, not classified as failures of the $SU(3)$ interpretation.

4.3. One generation of chiral matter

Under the Lie algebra

$$\mathfrak{g}_{\mathrm{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y, \tag{4.6}$$

one observed generation, written with conventional right-handed fields, transforms as

$$\begin{aligned} Q_L &: (\mathbf{3},\mathbf{2})_{1/6}, \quad u_R : (\mathbf{3},\mathbf{1})_{2/3}, \quad d_R : (\mathbf{3},\mathbf{1})_{-1/3}, \\ L_L &: (\mathbf{1},\mathbf{2})_{-1/2}, \quad e_R : (\mathbf{1},\mathbf{1})_{-1}. \end{aligned} \tag{4.7}$$

An optional right-handed neutrino transforms as $(\mathbf{1},\mathbf{1})_0$. The scalar is

$$H : (\mathbf{1},\mathbf{2})_{1/2}. \tag{4.8}$$

Electric charges follow from $Q = T^3 + Y$, as derived in equation (3.8).

4.4. Anomaly closure

For anomaly calculations, express every fermion as left-handed. The conjugates u^c, d^c, e^c then have hypercharges $(-2/3, 1/3, 1)$. The mixed and cubic sums are

$$[SU(3)]^2U(1): \quad 2\left(\frac{1}{2}\right)\left(\frac{1}{6}\right)+\left(\frac{1}{2}\right)\left(-\frac{2}{3}\right)+\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)=0, \tag{4.9}$$

$$[SU(2)]^2U(1): \quad 3\left(\frac{1}{2}\right)\left(\frac{1}{6}\right)+\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)=0, \tag{4.10}$$

$$[\mathrm{grav}]^2U(1): \quad 6\left(\frac{1}{6}\right)+3\left(-\frac{2}{3}\right)+3\left(\frac{1}{3}\right)+2\left(-\frac{1}{2}\right)+1=0, \tag{4.11}$$

$$[U(1)]^3: \quad 6\left(\frac{1}{6}\right)^3+3\left(-\frac{2}{3}\right)^3+3\left(\frac{1}{3}\right)^3+2\left(-\frac{1}{2}\right)^3+1^3=0. \tag{4.12}$$

The number of left-handed weak doublets is $3+1=4$, so the global $SU(2)$ anomaly is absent. The purely colour cubic anomaly cancels between the two triplet components of Q_L and the conjugate singlets.

DERIVED RESULT 4.2 — ONE-GENERATION ANOMALY CLOSURE

The representation set equation (4.7) is locally and globally anomaly-free. This is a stringent consistency certificate for the representation pattern. It does not alone prove that the framework has uniquely selected the pattern; that stronger task is transferred to the branching analysis.

4.5. The complete renormalizable Lagrangian

With three generations understood, the renormalizable Standard Model Lagrangian can be organized as

$$\begin{aligned}\mathscr{L}_{\mathrm{SM}} = &-\frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu}-\frac{1}{4}W^i_{\mu\nu}W^{i\mu\nu}-\frac{1}{4}B_{\mu\nu}B^{\mu\nu}\\ &+\sum_\psi \bar\psi i\gamma^\mu D_\mu\psi+(D_\mu H)^\dagger(D^\mu H)-V(H)\\ &-\left(\bar Q_L Y_d H d_R+\bar Q_L Y_u\widetilde H u_R+\bar L_L Y_e H e_R+\text{h.c.}\right)\\ &+\frac{\theta_s g_s^2}{32\pi^2}G^a_{\mu\nu}\widetilde G^{a\mu\nu},\end{aligned} \tag{4.13}$$

where $\widetilde H = i\tau^2 H^*$. Neutrino masses require an extension: a dimension-five Majorana operator, right-handed neutrinos with Dirac or Majorana terms, or another specified mechanism.

After electroweak reduction,

$$M_u=\frac{v}{\sqrt2}Y_u, \qquad M_d=\frac{v}{\sqrt2}Y_d, \qquad M_e=\frac{v}{\sqrt2}Y_e. \tag{4.14}$$

Biunitary diagonalization of M_u and M_d leaves their relative left-handed rotation in the charged current:

$$V_{\mathrm{CKM}}=U_{uL}^\dagger U_{dL}. \tag{4.15}$$

An analogous construction gives lepton mixing once the neutrino mass mechanism is chosen.

4.6. Lie algebra versus global gauge group

Local interactions determine the algebra equation (4.6), but not by themselves the global group. Matter representations are compatible with quotients of

$$SU(3)_c \times SU(2)_L \times U(1)_Y \tag{4.16}$$

by discrete central subgroups, including a commonly studied $\mathbb{Z}_6$ quotient. This distinction affects allowed bundles, line operators, monopoles, and possible unification embeddings. The programme therefore carries up to three global interpretations:

1. the direct product;
2. a quotient by a subgroup of $\mathbb{Z}_6$;
3. a global structure inherited uniquely from a simply connected unified group.

4.7. Framework obligations concentrated

At this stage the image's Standard Model box has a complete conventional action and an exact consistency ledger. The internal framework must now determine which parts are outputs rather than inputs:

1. the gauge algebra and global group;
2. chiral representations and the number of generations;
3. the scalar or alternative symmetry-reduction sector;
4. dimensionless gauge, Yukawa, and scalar couplings at a declared scale;
5. the strong topological angle or a mechanism controlling it;
6. neutrino masses;
7. the map from framework states to scattering and bound-state observables.

CONDITIONAL RESULT 4.3 — STANDARD MODEL CLOSURE UNDER SUPPLIED PHYSICAL PREMISES

Once equations (4.7) and (4.8) and the renormalizable local action are supplied, the strong, weak, electromagnetic, mass, mixing, and anomaly structures follow coherently. The deeper derivation problem is therefore sharply localized to selection of representations, vacuum structure, parameters, and global topology.

Electronuclear unification and representation branching

The image labels its upper particle branch *electronuclear force (Grand Unified Theory).* The mathematical content required by that label is a group G_{U} with a representation and vacuum structure that recover the Standard Model sectors, fix relative generator normalizations, and supply a consistent scale evolution. The corpus's finite carrier and block-splitting constructions make this a natural place to compare several internally coherent interpretations.

5.1. The minimal $SU(5)$ branching calculation

Let the defining space split as

$$\mathbb{C}^5 = \mathbb{C}^3_c \oplus \mathbb{C}^2_L. \tag{5.1}$$

The traceless diagonal operator commuting with $SU(3)_c \times SU(2)_L$ is unique up to scale. Requiring the conventional Higgs-doublet hypercharge fixes

$$Y = \operatorname{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right), \qquad \operatorname{tr} Y = 0. \tag{5.2}$$

Therefore

$$\begin{aligned} \mathbf{5} &\longrightarrow (\mathbf{3}, \mathbf{1})_{-1/3} \oplus (\mathbf{1}, \mathbf{2})_{1/2}, \\ \overline{\mathbf{5}} &\longrightarrow (\overline{\mathbf{3}}, \mathbf{1})_{1/3} \oplus (\mathbf{1}, \mathbf{2})_{-1/2}. \end{aligned} \tag{5.3}$$

The antisymmetric representation is $\mathbf{10} = \Lambda^2\mathbf{5}$. Applying

$$\Lambda^2(V \oplus W) = \Lambda^2 V \oplus (V \otimes W) \oplus \Lambda^2 W \tag{5.4}$$

to equation (5.1) gives

$$\mathbf{10} \longrightarrow (\overline{\mathbf{3}}, \mathbf{1})_{-2/3} \oplus (\mathbf{3}, \mathbf{2})_{1/6} \oplus (\mathbf{1}, \mathbf{1})_{1}. \tag{5.5}$$

Consequently

$$\overline{\mathbf{5}} \oplus \mathbf{10} \longleftrightarrow (d^c, L) \oplus (u^c, Q, e^c), \tag{5.6}$$

which is exactly one anomaly-free Standard Model generation without a right-handed neutrino (Georgi and Glashow, 1974).

The adjoint branches as

$$\begin{aligned}\mathbf{24} \longrightarrow\ & (\mathbf{8},\mathbf{1})_0 \oplus (\mathbf{1},\mathbf{3})_0 \oplus (\mathbf{1},\mathbf{1})_0 \\ & \oplus (\mathbf{3},\mathbf{2})_{-5/6} \oplus (\overline{\mathbf{3}},\mathbf{2})_{5/6}.\end{aligned} \tag{5.7}$$

The first three terms contain the Standard Model gauge connections; the final two are additional unified gauge directions.

DERIVED RESULT 5.1 — EXACT $SU(5)$ CHARGE RECONSTRUCTION

The five-direction $3+2$ block split, tracelessness, and exterior-square algebra generate the complete conventional hypercharge pattern for one generation. This is an exact representation-theoretic derivation conditional on choosing the $SU(5)$ realization and identifying its $3+2$ stabilizer with colour and weak isospin.

5.2. Generator normalization and the high-scale weak angle

In the fundamental representation,

$$\mathrm{tr}(Y^2) = 3\left(\frac{1}{9}\right) + 2\left(\frac{1}{4}\right) = \frac{5}{6}. \tag{5.8}$$

The canonically normalized generator satisfying $\mathrm{tr}(T_Y^2) = 1/2$ is

$$T_Y = \sqrt{\frac{3}{5}}\,Y. \tag{5.9}$$

Thus the GUT-normalized Abelian coupling is

$$g_1 = \sqrt{\frac{5}{3}}\,g'. \tag{5.10}$$

If $g_1 = g_2 = g_U$ at the symmetry-restoration scale, then $g'^2 = (3/5)g_U^2$, and

$$\sin^2\theta_W = \frac{g'^2}{g_2^2 + g'^2} = \frac{3}{8} \tag{5.11}$$

at that scale. Renormalization-group evolution and threshold corrections are required before comparison with a low-energy measurement.

5.3. Larger interpretation I: $SO(10)$

A chiral spinor of $SO(10)$ contains

$$\mathbf{16} \longrightarrow \mathbf{10}_{-1} \oplus \overline{\mathbf{5}}_3 \oplus \mathbf{1}_{-5} \tag{5.12}$$

under $SU(5) \times U(1)_\chi$, up to an overall sign convention for the $U(1)_\chi$ charge (Fritzsch and Minkowski, 1975). The singlet has precisely the Standard Model quantum numbers of a right-handed-neutrino conjugate.

This interpretation has three structural advantages:

1. one irreducible chiral multiplet contains the full generation including ν^c;
2. anomaly cancellation is inherited from the unified representation;
3. Majorana masses and a seesaw realization become available after specifying the symmetry-reduction sector.

It does not by group theory alone determine the Yukawa matrices, vacuum chain, or number of replicated spinors.

5.4. Larger interpretation II: E_6

The fundamental chiral representation of E_6 branches as

$$\mathbf{27} \longrightarrow \mathbf{16}_1 \oplus \mathbf{10}_{-2} \oplus \mathbf{1}_4 \tag{5.13}$$

under $SO(10) \times U(1)_\psi$, again with convention-dependent overall charge sign (Gürsey et al., 1976). It contains one $SO(10)$ family together with a vector multiplet and a singlet. The additional states can participate in symmetry reduction or acquire large masses, but the mass mechanism must be derived rather than assumed.

5.5. Carrier ancestry through E_8

The corpus frequently uses an E_8-based carrier. Two exact maximal-subgroup branchings organize possible physical interpretations:

$$\mathbf{248}\big|_{E_6\times SU(3)} = (\mathbf{78},\mathbf{1}) \oplus (\mathbf{1},\mathbf{8}) \oplus (\mathbf{27},\mathbf{3}) \oplus (\overline{\mathbf{27}},\overline{\mathbf{3}}), \tag{5.14}$$

$$\mathbf{248}\big|_{SO(10)\times SU(4)} = (\mathbf{45},\mathbf{1}) \oplus (\mathbf{1},\mathbf{15}) \oplus (\mathbf{16},\mathbf{4}) \oplus (\overline{\mathbf{16}},\overline{\mathbf{4}}) \oplus (\mathbf{10},\mathbf{6}). \tag{5.15}$$

The factor $\mathbf{3}$ in equation (5.14) makes a threefold family interpretation mathematically visible. The simultaneous conjugate sector means that chirality is not selected by the adjoint branching alone. A projection, boundary condition, index, grading, or other asymmetric realization must specify which low-energy modes persist. Representation-theoretic constraints on placing chiral gravity and matter in a bare E_8 adjoint have been analysed in the literature (Distler and Garibaldi, 2010); the present construction therefore treats the carrier as richer data than an unqualified adjoint-as-particle-list identification.

Conditional result 5.2 — three retained unification interpretations

The current premises support three nested interpretations: $SU(5)$ as the minimal charge-branching algebra; $SO(10)$ as one-family closure including a neutral state; and E_6 or an $E_8 \supset E_6 \times SU(3)$ ancestry as a route to additional states and family structure. These are not arbitrary alternatives: each contains the preceding Standard Model representation structure. Chirality, symmetry reduction, and the mass spectrum are the discriminating obligations.

5.6. Scale evolution as a consistency equation

At one loop the inverse gauge couplings obey

$$\alpha_i^{-1}(\mu) = \alpha_i^{-1}(\mu_0) - \frac{b_i}{2\pi} \ln \frac{\mu}{\mu_0}. \tag{5.16}$$

For the Standard Model with $g_1 = \sqrt{5/3}\, g'$,

$$(b_1, b_2, b_3) = \left(\frac{41}{10}, -\frac{19}{6}, -7\right). \tag{5.17}$$

Pairwise equality occurs at

$$\ln \frac{\mu_{ij}}{\mu_0} = \frac{2\pi \left[\alpha_i^{-1}(\mu_0) - \alpha_j^{-1}(\mu_0)\right]}{b_i - b_j}. \tag{5.18}$$

Exact single-scale meeting is therefore a calculable constraint on particle content and thresholds, not a consequence of embedding the algebra alone.

5.7. Physical discriminants

Any retained interpretation must eventually calculate:

1. the complete symmetry-reduction chain and vacuum stability;
2. threshold-corrected coupling evolution;
3. masses and quantum numbers of additional gauge and matter states;
4. baryon-number-violating operators and proton-decay channels;
5. neutrino masses and mixings;
6. topological defects implied by the global breaking pattern;
7. the mechanism selecting chirality and the observed number of light families.

The exact branchings already narrow the task sharply. The internal framework need not invent particle labels one by one; it must select a carrier reduction and persistence rule whose surviving representation content supplies them.

Spacetime curvature and the cosmological reduction

6.1. Metric, connection, and curvature

Let $g_{\mu\nu}$ be a Lorentzian metric and ∇ its torsion-free metric-compatible connection. The Christoffel symbols are

$$\Gamma^{\rho}{}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} \left(\partial_{\mu} g_{\sigma\nu} + \partial_{\nu} g_{\sigma\mu} - \partial_{\sigma} g_{\mu\nu} \right), \tag{6.1}$$

and the curvature is

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_{\mu}\Gamma^{\rho}{}_{\nu\sigma} - \partial_{\nu}\Gamma^{\rho}{}_{\mu\sigma} + \Gamma^{\rho}{}_{\mu\lambda}\Gamma^{\lambda}{}_{\nu\sigma} - \Gamma^{\rho}{}_{\nu\lambda}\Gamma^{\lambda}{}_{\mu\sigma}. \tag{6.2}$$

Contracting gives $R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$, $R = g^{\mu\nu} R_{\mu\nu}$, and the Einstein tensor

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}. \tag{6.3}$$

The differential Bianchi identity implies

$$\nabla^{\mu} G_{\mu\nu} = 0. \tag{6.4}$$

6.2. Variation of the gravitational action

Take

$$S[g,\Psi] = \frac{1}{2\kappa} \int_M \mathrm{d}^4x \sqrt{-g}\,(R - 2\Lambda) + S_{\mathrm{m}}[g,\Psi], \qquad \kappa = \frac{8\pi G}{c^4}. \tag{6.5}$$

The matter stress–energy tensor is defined by

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_{\mathrm{m}}}{\delta g^{\mu\nu}}. \tag{6.6}$$

Using

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\, g_{\mu\nu}\delta g^{\mu\nu} \tag{6.7}$$

and the Palatini identity, the variation of $\sqrt{-g}R$ is a bulk term $\sqrt{-g}G_{\mu\nu}\delta g^{\mu\nu}$ plus a boundary divergence. With an appropriate boundary term or fixed boundary data,

$$\delta S = \frac{1}{2}\int_M \mathrm{d}^4x\sqrt{-g}\left[\frac{1}{\kappa}(G_{\mu\nu} + \Lambda g_{\mu\nu}) - T_{\mu\nu}\right]\delta g^{\mu\nu}. \tag{6.8}$$

Stationarity for arbitrary interior variations yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \tag{6.9}$$

the metric field equation of general relativity (Einstein, 1916).

Combining equations (6.4) and (6.9) gives

$$\nabla^\mu T_{\mu\nu} = 0. \tag{6.10}$$

Thus the matter–curvature cross-edge is not an optional verbal association: the same diffeomorphism-compatible action produces both the source tensor and its covariant conservation.

Derived result 6.1 — the curvature branch

Given a Lorentzian metric, the Einstein–Hilbert action, and a diffeomorphism-covariant matter action, variation gives the Einstein equation and the Bianchi identity gives covariant stress–energy conservation. Electromagnetism couples through equation (2.15); the remaining Standard Model fields couple through their covariant matter action.

6.3. First-order and carrier-compatible form

Introduce a coframe e^a and Lorentz connection $\omega^{ab} = -\omega^{ba}$, with

$$T^a = \mathrm{d}e^a + \omega^a{}_b \wedge e^b, \qquad R^{ab} = \mathrm{d}\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}. \tag{6.11}$$

The Einstein–Cartan action with cosmological term is

$$S_{\mathrm{EC}} = \frac{1}{4\kappa}\int \epsilon_{abcd}\left(e^a \wedge e^b \wedge R^{cd} - \frac{\Lambda}{6}e^a \wedge e^b \wedge e^c \wedge e^d\right). \tag{6.12}$$

Independent variation of e and ω yields a gravitational equation and a torsion equation. In vacuum or without intrinsic spin, the latter commonly gives $T^a = 0$; spin-carrying matter can source torsion.

The corpus's connection-square proposal has a standard conditional route. For a de Sitter or anti-de Sitter connection whose broken components contain ω^{ab} and e^a/ℓ, the curvature splits schematically as

$$\mathcal{F}^{ab} = R^{ab} \pm \ell^{-2}e^a \wedge e^b, \qquad \mathcal{F}^{a4} \propto T^a. \tag{6.13}$$

A projected quadratic action in $\mathscr{F}$ expands into a topological curvature square, an Einstein–Cartan term, and a cosmological term (MacDowell and Mansouri, 1977). This establishes a concrete bridge from a larger connection to curvature dynamics once the projection, invariant form, and symmetry-reduction scale are specified.

6.4. Homogeneity and isotropy

A spatially homogeneous and isotropic metric can be written

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2(t)\left[\frac{\mathrm{d}r^2}{1-kr^2} + r^2\mathrm{d}\Omega^2\right], \tag{6.14}$$

where k is the normalized spatial-curvature parameter and $H = \dot{a}/a$. For a perfect fluid,

$$T_{\mu\nu} = \left(\rho + \frac{p}{c^2}\right)u_\mu u_\nu + p g_{\mu\nu}. \tag{6.15}$$

Substitution into equation (6.9) gives

$$H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \tag{6.16}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}. \tag{6.17}$$

Covariant conservation reduces to

$$\dot{\rho} + 3H\left(\rho + \frac{p}{c^2}\right) = 0. \tag{6.18}$$

These are the background equations underlying the standard cosmological model (Friedmann, 1922).

For a component with constant $w = p/(\rho c^2)$, integration yields

$$\rho(a) = \rho_0 a^{-3(1+w)}. \tag{6.19}$$

Thus radiation scales as a^{-4}, pressureless matter as a^{-3}, and a cosmological constant as a^0.

Derived result 6.2 — curvature to background cosmology

The line from spacetime curvature to the standard cosmological background is a symmetry reduction. It follows from the gravitational field equation, the FLRW ansatz, and a perfect-fluid source. No separate cosmological force is introduced.

6.5. What a complete cosmological model additionally contains

Background expansion alone is not the complete standard cosmological model. Comparison with observations requires

1. linear scalar, vector, and tensor perturbations;
2. a gauge choice or gauge-invariant perturbation variables;
3. photon, baryon, neutrino, and dark-sector kinetic equations;
4. recombination and radiative transfer;
5. nonlinear structure formation or calibrated effective modeling;
6. source populations, selection functions, covariance, and likelihoods.

The framework's algebraic density dictionary may specify initial or background parameter relations, but a physical derivation must propagate them through this full observation map.

6.6. Three retained geometric interpretations

The current corpus supports three compatible levels:

Metric effective interpretation.
: The framework produces an effective metric and stress–energy at resolved scales; equation (6.5) is the leading derivative action.

First-order connection interpretation.
: Coframe and connection are independent carrier components; Einstein–Cartan dynamics arise after variation.

Broken unified-connection interpretation.
: Coframe and spin connection descend from a larger gauge connection through a projection such as equation (6.13).

The levels can be nested rather than mutually exclusive. Torsion, additional propagating modes, coefficient relations, and high-curvature behavior determine how far the nesting extends.

6.7. Residual obligations

The classical curvature-to-cosmology line is conditionally complete. Framework-specific closure requires:

1. deriving the gravitational action and invariant form from the carrier;
2. fixing the sign and magnitude of G and Λ;
3. deriving which connection components propagate;

4. specifying matter coupling and torsion;
5. recovering the measured weak-field and wave regimes;
6. deriving cosmological perturbations and a prospective observable map.

CHAPTER 7

Quantum gravity and the meaning of complete closure

7.1. The quantum-gravity interface

The quantum-gravity box is not one additional force sector parallel to $SU(3)$ or $U(1)$. It identifies the sector of the theory in which gravitational degrees of freedom and quantum states, amplitudes, or observables coexist consistently. At minimum, its microscopic realization must provide

$$\mathfrak{Q}_{\mathrm{G}} = (\mathcal{S}, \mathcal{A}, \mathcal{C}, \mathcal{O}, \mathcal{D}, \mathcal{L}_{\mathrm{cl}}, \mathcal{L}_{\mathrm{QFT}}), \tag{7.1}$$

where $\mathcal{S}$ is a state space, $\mathcal{A}$ an observable algebra, $\mathcal{C}$ the gauge or diffeomorphism constraints, $\mathcal{O}$ an evolution or amplitude rule, $\mathcal{D}$ a physical inner product or probability prescription, and the final two entries are controlled classical-gravity and quantum-field-theory limits.

7.2. Low-energy quantum gravity is already a controlled limit

Write

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \kappa h_{\mu\nu} \tag{7.2}$$

around a background $\bar{g}$. The most general local diffeomorphism-invariant effective action is ordered by derivatives:

$$\begin{aligned} S_{\mathrm{eff}} = \int \mathrm{d}^4x\sqrt{-g}\bigg[&\frac{1}{2\kappa}(R - 2\Lambda) + c_1 R^2 + c_2 R_{\mu\nu}R^{\mu\nu} \\ &+ c_3 R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} + \cdots\bigg]. \end{aligned} \tag{7.3}$$

At energies $E \ll M_{\mathrm{Pl}}$, higher-curvature terms are suppressed by powers of E/M_{Pl}. Loop effects can therefore produce definite low-energy corrections even though additional coefficients are required at successive orders (Donoghue, 1994).

Established result 7.1 — effective quantum-gravity regime

General relativity admits a systematic low-energy quantum effective-field-theory treatment. Perturbative nonrenormalizability means that infinitely many higher-order operators enter toward arbitrarily high energy; it does not mean that the low-energy quantum theory lacks predictive content.

7.3. Canonical constraint route

Foliating spacetime into spatial metrics h_{ij}, lapse N, and shift N^i leads schematically to

$$S = \int \mathrm{d}t \int_\Sigma \mathrm{d}^3x \left(\pi^{ij}\dot{h}_{ij} - N\mathscr{H}_\perp - N^i\mathscr{H}_i\right). \tag{7.4}$$

The lapse and shift act as multipliers, so physical configurations satisfy

$$\mathscr{H}_\perp = 0, \qquad \mathscr{H}_i = 0. \tag{7.5}$$

A formal canonical quantization replaces momenta by functional derivatives and imposes

$$\widehat{\mathscr{H}_\perp}\Psi[h,\varphi] = 0, \qquad \widehat{\mathscr{H}_i}\Psi[h,\varphi] = 0. \tag{7.6}$$

The absence of an external time parameter is not an inconsistency: it reflects reparametrization invariance. A physical clock must be relational, selected from correlations among degrees of freedom.

This route fits the corpus's traversal language naturally. A record variable can serve as an internal clock if it is monotonic on the state sector considered and if the conditional dynamics of the remaining variables is unitary or otherwise probabilistically consistent. Those conditions must be demonstrated for the selected clock rather than assumed globally.

7.4. Covariant amplitude route

A second class uses amplitudes over geometries and matter histories,

$$Z[\mathscr{B}] = \int_{\mathscr{B}} \mathscr{D}g\,\mathscr{D}\Psi\, e^{iS[g,\Psi]/\hbar}, \tag{7.7}$$

with boundary data $\mathscr{B}$, gauge fixing, ghosts or an equivalent quotient construction, a specified measure, and a contour or convergence prescription. The framework's additive-action phase supplies a conditional reason for the exponential character:

$$\chi(S_1 + S_2) = \chi(S_1)\chi(S_2), \quad \chi : \mathbb{R} \to U(1), \quad \chi(S) = e^{iS/\hbar}, \tag{7.8}$$

where continuity and the unitary target are explicit premises. This derives the phase form, not the measure, gravitational action, or physical state space.

7.5. Discrete carrier route

A third class treats the carrier as finite or combinatorial at a fundamental level and recovers continuum geometry from large-scale states. Let Γ be a labelled complex and let

$$Z_\Gamma = \sum_{\ell \in \mathscr{L}(\Gamma)} \prod_f A_f(\ell) \prod_e A_e(\ell) \prod_v A_v(\ell) \tag{7.9}$$

be a generic local amplitude. A successful realization must establish:

1. independence from arbitrary subdivisions or a controlled renormalization flow;
2. a semiclassical sector whose correlation functions reproduce a Lorentzian metric;
3. the correct massless spin-two propagation and nonlinear coupling;
4. matter representations and their chiral dynamics;
5. probabilities for boundary records.

The corpus's finite root systems, traversals, and retained records make this interpretation mathematically natural, while leaving the amplitude and continuum-limit theorem to be selected.

7.6. The required correspondence equations

Every interpretation must reproduce a semiclassical relation of the form

$$G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = \kappa \langle \widehat{T}_{\mu\nu} \rangle_{\text{ren}} + \Delta_{\mu\nu}, \tag{7.10}$$

where $\Delta_{\mu\nu}$ contains controlled higher-curvature or quantum-state corrections. It must also recover ordinary quantum fields on backgrounds over the domain where backreaction is negligible.

Black-hole thermodynamics provides a cross-check because horizon entropy and radiation couple geometry, quantum fields, and statistical mechanics. The area law and Hawking temperature are

$$S_{\text{BH}} = \frac{k_{\text{B}} c^3 A}{4G\hbar}, \qquad T_{\text{H}} = \frac{\hbar \kappa_H}{2\pi k_{\text{B}} c}, \tag{7.11}$$

for horizon area A and surface gravity κ_H (Bekenstein, 1973; Hawking, 1975). A carrier interpretation should count or otherwise explain the relevant states without changing these limits outside a quantitatively specified correction regime.

7.7. Three retained quantum-gravity interpretations

Continuum effective interpretation.
The carrier fixes the coefficients and matter content of equation (7.3); unknown ultraviolet structure is encoded in higher operators.

Constrained connection interpretation.
A generalized connection is quantized, and physical states solve a closed constraint algebra; geometry and matter are relational observables.

Finite traversal interpretation.
Fundamental states are labelled carrier configurations and histories; continuum fields emerge from coarse-grained amplitudes such as equation (7.9).

These may describe nested scales rather than exclusive theories. The discriminants are the microscopic state count, ultraviolet amplitudes, correction operators, and continuum-limit map.

Conditional result 7.2 — what the framework already supplies

The framework supplies a configuration carrier, ordered composition, additive action, complex phase, bounded records, and observer-relative resolution. Those ingredients cover several entries of equation (7.1). The three retained microscopic realizations differ in their selected physical state space, constraint or amplitude rule, inner product, and route to the classical and quantum-field limits; those differences define their discriminating calculations.

7.8. Quantum gravity within the theory of everything

Quantum gravity alone is not the top box. Complete closure must also derive or contain

1. the Standard Model representations and interactions;
2. symmetry reduction and dimensional scales;
3. flavour, neutrino, and vacuum structure;
4. cosmological states and perturbations;
5. operational observables, probabilities, and records;
6. a finite parameter ledger or an explicit explanation of remaining environmental data;
7. a common domain in which all limits agree.

Define a closure graph $\mathscr{G}$ whose vertices are mathematical sectors and whose edges are realization maps or controlled limits. A theory-of-everything claim is complete only if every empirical vertex is reachable from the foundational data by a path whose assumptions are recorded:

$$\forall p \in \mathscr{P}_{\mathrm{emp}}, \quad \exists \gamma \,:\, \mathscr{F}_{\mathrm{root}} \rightsquigarrow p, \quad A(\gamma) \text{ explicit.} \tag{7.12}$$

The present dossier uses equation (7.12) to audit the theory's closure graph. Reachability and compatibility establish the common architecture; uniqueness of every microscopic realization or numerical selector is a stronger local question and is recorded separately.

CHAPTER 8

Internal synthesis of the framework

8.1. Three derivation ladders

The corpus becomes most coherent when its tools are organized into three ladders that meet at physical observables.

Algebraic ladder.
Finite carrier → admissible subalgebras → representations → charge operators → gauge and matter sectors.

Geometric ladder.
Connection → curvature → invariant action → field equations → classical backgrounds and perturbations.

Operational ladder.
Distinction → ordered action → phase or weight → channel → resolved record and empirical likelihood.

A physical theory is internally described only where the ladders commute. For example, the algebraic ladder can identify a photon generator, the geometric ladder can supply its curvature and stress–energy, and the operational ladder can define a detector record. If two paths from the carrier to the same observable disagree, that disagreement is a calculable discriminator among interpretations.

8.2. The commuting-realization condition

Let $\mathscr{F}$ denote the framework carrier, $\mathscr{M}$ a mathematical intermediate description, and $\mathscr{P}$ a physical observable model. Suppose two routes are available:

$$\mathscr{F} \xrightarrow{r_1} \mathscr{M}_1 \xrightarrow{s_1} \mathscr{P}, \qquad \mathscr{F} \xrightarrow{r_2} \mathscr{M}_2 \xrightarrow{s_2} \mathscr{P}. \tag{8.1}$$

Internal self-description requires

$$s_1 \circ r_1 \simeq s_2 \circ r_2 \tag{8.2}$$

on the common observable domain, where $\simeq$ means equality up to declared gauge, convention, approximation, or statistical tolerance.

Internal result 8.1 — loose threads become commutativity tests

Every apparently separate interpretive bridge can be rewritten as a comparison of two routes to a common observable. The framework can therefore constrain itself internally: representation branching, connection dynamics, and measurement readout must agree where they describe the same charge, mass, amplitude, or record.

8.3. Particle-sector convergence

The exact algebraic chain presently available is

$$E_8 \supset E_6 \times SU(3) \supset SO(10) \times U(1) \times SU(3) \supset SU(5) \times U(1)^2 \times SU(3), \tag{8.3}$$

followed by

$$SU(5) \supset SU(3)_c \times SU(2)_L \times U(1)_Y. \tag{8.4}$$

The branchings in Chapter 5 reconstruct the Standard Model charge pattern and make a family index visible. The persistence rule must now do three things simultaneously:

1. select chiral rather than paired low-energy matter;
2. retain exactly the required light representations;
3. give masses to complementary states without breaking $U(1)_{\mathrm{EM}}$ or introducing anomalies.

Three mechanisms remain mathematically coherent at this stage:

1. an index or topological projection on an internal space;
2. boundary or orbifold conditions selecting one chirality;
3. a dynamical grading/condensate whose kernel is the retained chiral sector.

They will be compared through the net chiral index, scalar spectrum, anomaly inflow, and threshold corrections.

8.4. Geometric-sector convergence

The carrier's connection interpretation and the gravitational square converge if there exists an invariant projection P such that

$$S[\mathscr{A}] = \int \langle P\mathscr{F} \wedge P\mathscr{F} \rangle \tag{8.5}$$

reduces to equation (6.12) plus topological and controlled correction terms. This requires:

1. a real form compatible with Lorentzian signature;
2. a nondegenerate coframe sector;
3. the correct sign of the effective Newton constant;
4. a stable symmetry-reduction vacuum;
5. a matter coupling whose stress–energy agrees with equation (6.6).

The coefficient correspondence identified in the earlier corpus is therefore one equation within a larger commutative square:

$$\begin{tikzcd} \boxed{\begin{array}{c}\text{carrier connection}\\ \text{and invariant}\end{array}} \arrow[r, "\text{projection}"] \arrow[d, "\text{direct effective map}"'] & \boxed{\begin{array}{c}\text{coframe–connection}\\ \text{action}\end{array}} \arrow[d, "\text{variation}"] \\ \boxed{\begin{array}{c}\text{weak-field and}\\ \text{cosmological records}\end{array}} & \boxed{\begin{array}{c}\text{Einstein–Cartan}\\ \text{field equations}\end{array}} \arrow[l, "\text{solution/readout}"] \end{tikzcd} \tag{8.6}$$

Agreement of the lower and diagonal routes is the required certificate.

8.5. Operational convergence

For additive action S and continuous unitary composition,

$$\chi(S) = e^{iS/\hbar} \tag{8.7}$$

follows as the continuous character of $(\mathbb{R}, +)$. A measurement channel then maps a prepared state to an outcome record,

$$\rho \xrightarrow{\mathscr{E}} \rho' \xrightarrow{\{M_y\}} p(y) = \mathrm{Tr}(\rho' M_y), \tag{8.8}$$

where $\mathscr{E}$ is completely positive and trace preserving and the effects satisfy $M_y \geq 0$, $\sum_y M_y = I$. The framework's observer-relative resolution can be represented by a coarse-graining channel

$$\mathscr{C} : \{y\} \longrightarrow \{\bar{y}\}, \qquad p(\bar{y}) = \sum_{y \in \bar{y}} p(y). \tag{8.9}$$

This supplies a precise realization of prepared state, propagation, resolution, and retained record. A further interpretation is needed only if the framework intends to specify ontology beyond operational probabilities.

8.6. Scale and parameter closure

Pure representation theory fixes rational charges and multiplicities but does not fix dimensionful scales. A scale can enter through one of three routes:

1. a primitive dimensional constant in the carrier action;
2. dimensional transmutation through running couplings;
3. a state-dependent or boundary-dependent vacuum scale.

Whichever route is selected must produce the same measured scale when inferred independently from, for example, G_F, particle masses, gravitational strength, and cosmological observables.

Define the parameter map

$$\Pi : \Theta_{\text{carrier}} \longrightarrow \left\{g_s, g, g', v, \lambda, Y_f, G, \Lambda, \ldots\right\}. \tag{8.10}$$

Closure does not require every low-energy parameter to be a simple algebraic constant. It does require the number of independent inputs and their renormalization conditions to be explicit.

8.7. Current synthesis

The derivation programme has now established the following chain conditional on named realization data:

$$\begin{aligned}
&\text{carrier subgroup chain} && \Longrightarrow && \text{Standard Model charges and anomaly closure,} \\
&SU(2)_L \times U(1)_Y + \langle H \rangle && \Longrightarrow && U(1)_{\text{EM}} + W^{\pm} + Z, \\
&U(1)_{\text{EM}} \text{ connection} && \Longrightarrow && F \Longrightarrow (E_u, B_u), \\
&\text{metric/coframe action} && \Longrightarrow && \text{curvature dynamics} \Longrightarrow \text{FLRW background.}
\end{aligned} \tag{8.11}$$

The remaining threads are concentrated in the selection maps: chirality, vacuum structure, parameters, nonperturbative dynamics, quantum-gravitational state space, and prospective observation. None is treated as disconnected from the rest. Each is a missing side of a commuting diagram to be constrained by the already-derived branches.

CONDITIONAL SYNTHESIS 8.1 — SELF-DESCRIPTION FROM INTERSECTING ROUTES

The framework can describe itself internally if the algebraic, geometric, and operational ladders yield the same low-energy objects and records. The strongest next derivations are therefore those with multiple independent routes: photon charge and coupling, family chirality, the gravitational coefficient, vacuum scale, and cosmological perturbations.

CHAPTER 9

The framework-specific realization

The preceding chapters established the mathematics carried by each box in the supplied diagram. This chapter now performs the framework-specific composition in chronological order. Its purpose is not to place familiar names onto symbols. It is to show exactly where the internal construction forces a structure, where a physical bridge is still required, and how the same bridge must commute with every lower-energy reduction.

9.1. Stage 1: the retained fraction and the three weights

Let

$$\varphi^2 = \varphi + 1, \qquad r = \frac{1}{2\varphi}, \qquad u = 1 - r. \tag{9.1}$$

The self-relation of the two complementary shares is

$$(u + r)^2 = \underbrace{u^2}_{W_\mathrm{L}} + \underbrace{2ur}_{W_\mathrm{B}} + \underbrace{r^2}_{W_\mathrm{M}} = 1. \tag{9.2}$$

Thus

$$W_\mathrm{L} = \frac{15 - 5\sqrt{5}}{8}, \qquad W_\mathrm{B} = \frac{3\sqrt{5} - 5}{4}, \qquad W_\mathrm{M} = \frac{3 - \sqrt{5}}{8}, \tag{9.3}$$

and the exact ratios are

$$\frac{W_\mathrm{M}}{W_\mathrm{L}} = \frac{1}{5}, \qquad \frac{W_\mathrm{B}}{W_\mathrm{L}} = \frac{2}{\sqrt{5}}. \tag{9.4}$$

Equations (9.1)–(9.4) are algebraic. The names light, boundary and matter are a later physical interpretation. The distinction matters: the arithmetic survives even if the physical reading changes.

9.2. Stage 2: carrier selection

The later corpus formulates a carrier as a positive-definite, even, unimodular lattice. Positive-definite even unimodular lattices exist only in ranks divisible by eight, and the rank-eight example is unique up to isometry: the E_8 lattice with 240 roots (Conway and Sloane, 1999). Therefore

$$\left.\begin{array}{c}\text{positive definite}\\ \text{even}\\ \text{unimodular}\\ \text{minimal nonzero rank}\end{array}\right\} \implies L \cong E_8. \tag{9.5}$$

Carrier statement

The implication in equation (9.5) is standard lattice mathematics. Its application to physics is conditional on the self-description principles genuinely implying the four properties on the left. The dossier therefore uses E_8 as a conditionally derived carrier, not as an inserted particle list (Shields, 2026d).

The shortest nondegenerate closed root history is triangular. If roots α and β satisfy $\alpha \cdot \beta = -1$, then $\gamma = -(\alpha + \beta)$ is also a root and

$$\alpha + \beta + \gamma = 0, \qquad w(\alpha\beta\gamma) = r^3. \tag{9.6}$$

This supplies a precise combinatorial meaning for the recurrent factor r^3 once a primitive retained crossing is identified with a root step.

9.3. Stage 3: reduction as a conditional expectation

Let $\mathscr{A}$ be the carrier observable algebra and $\mathscr{B} \subset \mathscr{A}$ the readable algebra. The reduction is represented by a conditional expectation

$$\mathscr{C} : \mathscr{A} \longrightarrow \mathscr{B}, \qquad \mathscr{C}^2 = \mathscr{C}, \qquad \mathscr{C}(1) = 1, \qquad \mathscr{C}(b_1 a b_2) = b_1 \mathscr{C}(a) b_2. \tag{9.7}$$

Complete positivity prevents the reduction from turning positive observables into negative ones. The fixed algebra is exactly the readable sector. Conditional expectations onto operator subalgebras are standard objects; the physical reading is the framework's contribution (Tomiyama, 1957; Shields, 2026d).

For a root subsystem R_{d}, the explicit group-average realization is

$$\mathscr{C}(a) = \frac{1}{|W(R_{\mathrm{d}})|} \sum_{w \in W(R_{\mathrm{d}})} \mathrm{Ad}_w(a). \tag{9.8}$$

It is unital, idempotent, completely positive, bimodular over its fixed algebra, and equivariant under the normalizer of the averaging group.

9.4. Stage 4: why the decayed subsystem is A_3

The geometry block contains one clock mode and one closure mode. If its total rank is five, the spatial rank is three and the decayed rank inside the rank-eight carrier is also three. Exhaustive enumeration gives three Weyl-conjugacy classes of rank-three root subsystems:

Decayed subsystem	Roots	Orthogonal survivor	Survivor roots
A_1^3	6	$D_4 + A_1$	26
$A_2 + A_1$	8	A_5	30
A_3	12	D_5	40

The corpus then imposes two pre-existing internal criteria: the decayed subsystem must host a closed triangular history, and a single physical descent must be governed by one irreducible decay operator. The first criterion removes A_1^3; the second removes $A_2 + A_1$. Hence

$$R_{\mathrm{d}} = A_3, \qquad R_{\mathrm{d}}^{\perp} = D_5, \qquad \mathfrak{g}_{\mathrm{surv}} \cong \mathfrak{so}(10). \tag{9.9}$$

For A_3, $W(A_3) \cong S_4$ has order 24, and equation (9.8) has a five-dimensional fixed subspace (Shields, 2026i).

Finite certificate 9.1

The supplied reproducibility certificate records 2,667 distinct rank-three spans, the three classes above, the two selection tests, an order-24 average, and a fixed-space dimension of five. This certifies the finite enumeration and matrix identities. The physical content of the two selection criteria remains an explicit realization premise.

9.5. Stage 5: gauge fields and matter from the same split

Under $A_3 \times D_5 \cong SU(4) \times SO(10)$, the E_8 adjoint branches as

$$\mathbf{248} = (\mathbf{15}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{45}) \oplus (\mathbf{6}, \mathbf{10}) \oplus (\mathbf{4}, \mathbf{16}) \oplus (\overline{\mathbf{4}}, \overline{\mathbf{16}}). \tag{9.10}$$

The dimensions close exactly:

$$15 + 45 + 60 + 64 + 64 = 248. \tag{9.11}$$

Operations commuting with the average lie in the D_5 commutant and are read as gauge operations. Off-diagonal modules connect distinct decay sectors and are read as matter. Their noncommutation with a general positive decay generator supplies a candidate mass operator. This yields

$$\text{commutant} = \mathfrak{so}(10), \qquad \text{spinor matter} = \mathbf{16}, \qquad \text{vector sector} = \mathbf{10}. \tag{9.12}$$

The group theory in equation (9.10) is established; the forcing of the A_3 reduction is the framework-specific part (Fritzsch and Minkowski, 1975; Shields, 2026i).

The family index transforms in the permutation representation of S_4:

$$\mathbf{4}_{\mathrm{perm}} = \mathbf{1}_{\mathrm{inv}} \oplus \mathbf{3}_{\mathrm{std}}. \tag{9.13}$$

If matter is restricted to directions connecting distinct decay sectors, the invariant direction is not counted as a separate readable family and three non-invariant directions remain. Thus the family count is conditionally derived from the matter criterion; the $1+3$ character decomposition itself is exact.

9.6. Stage 6: one surviving five-block, two compatible decompositions

The five-dimensional fixed block admits two different decompositions because two different questions are being asked:

$$\mathbf{5} = \mathbf{3}_X \oplus \mathbf{1}_T \oplus \mathbf{1}_C, \quad \text{external distinction, clock, closure,} \tag{9.14}$$

$$\mathbf{5} = \mathbf{3}_{\mathrm{rev}} \oplus \mathbf{2}_{\mathrm{irr}}, \quad \text{reversible versus arrow-sensitive directions.} \tag{9.15}$$

The first gives a four-dimensional traversable spacetime together with non-traversable closure data. The opposite roles of $\mathbf{3}_X$ and $\mathbf{1}_T$ produce the Lorentzian form

$$\eta = P_X - P_T. \tag{9.16}$$

The second preserves $S(U(3) \times U(2))$, whose Lie algebra is

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1). \tag{9.17}$$

There is no conflict between equations (9.14) and (9.15): the first classifies geometric role and the second classifies reversibility (Shields, 2026a,f).

9.7. Stage 7: hypercharge and one generation

Let hypercharge be constant on the colour and weak blocks,

$$Y = \mathrm{diag}(a, a, a, b, b). \tag{9.18}$$

Unimodularity gives $3a + 2b = 0$. With the conventional weak normalization $b = 1/2$,

$$Y = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right). \tag{9.19}$$

The ratio $a/b = -2/3$ is forced by the $3+2$ split; the overall scale is a charge convention until the coupling normalization is fixed. The even exterior algebra of the five complex directions has dimension

$$\dim \Lambda^{\mathrm{even}}\mathbb{C}^5 = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5 = 16, \tag{9.20}$$

and its wedge weights give

$$\begin{aligned} Q &: (\mathbf{3},\mathbf{2})_{1/6}, \quad & u^c &: (\bar{\mathbf{3}},\mathbf{1})_{-2/3}, \quad & d^c &: (\bar{\mathbf{3}},\mathbf{1})_{1/3}, \\ L &: (\mathbf{1},\mathbf{2})_{-1/2}, \quad & e^c &: (\mathbf{1},\mathbf{1})_{1}, \quad & \nu^c &: (\mathbf{1},\mathbf{1})_{0}. \end{aligned} \tag{9.21}$$

Thus all electric charges follow from $Q = T_3 + Y$. The anomaly sums vanish exactly, as shown in Chapter 4.

The same normalization gives

$$\operatorname{tr} Y^2 = \frac{5}{6}, \qquad \operatorname{tr} T_3^2 = \frac{1}{2}, \qquad k_Y = \frac{5}{3}, \qquad \sin^2\theta_W\big|_{\mathrm{unif}} = \frac{3}{8}. \tag{9.22}$$

This is the standard unified normalization. What the framework adds is a proposed derivation of the $3+2$ breaking direction from reversibility rather than its selection as an independent vacuum input.

9.8. Stage 8: electroweak and electromagnetic descent

Once a weak doublet order parameter with $Y = 1/2$ has a nonzero vacuum value, Chapter 3 proves

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\mathrm{EM}}, \qquad Q = T_3 + Y, \tag{9.23}$$

and derives the massless photon, the massive $W^\pm$ and Z, and $e = g\sin\theta_W = g'\cos\theta_W$. The framework currently admits three readings of that order parameter:

1. a fundamental scalar in the vector **10** of $SO(10)$;
2. a composite bilinear of off-diagonal carrier modes;
3. a component of the connection or decay field whose vacuum value selects the $3+2$ stabilizer.

All three produce the same tree-level reduction. They diverge in scalar self-couplings, additional states and high-energy form factors, so those quantities select between them.

The surviving Abelian curvature F then splits relative to an observer into the electric and magnetic fields. Electricity and magnetism are therefore not independent substances in the realization: they are the two observer-dependent projections of one $U(1)$ curvature, as proved in Chapter 2.

9.9. Stage 9: gravity and cosmology from the weighted square

For a larger connection with curvature $\mathscr{F}$ and realized area two-form $E = e \wedge e$, define

$$\mathscr{D} = u\,\mathscr{F} + r\,E. \tag{9.24}$$

Then

$$\mathscr{D} \wedge \mathscr{D} = W_{\mathrm{L}}\,\mathscr{F} \wedge \mathscr{F} + W_{\mathrm{B}}\,\mathscr{F} \wedge E + W_{\mathrm{M}}\,E \wedge E. \tag{9.25}$$

After the cluster projection and an invariant trace, these are respectively a curvature-square term, an Einstein-Cartan cross-term and a cosmological volume term. This is the MacDowell-Mansouri pattern (MacDowell and Mansouri, 1977; Shields, 2026f). Independent comparison of the last two coefficients gives the same carrier-unit value

$$\Lambda_{\text{carrier}} = \frac{3r}{u} = \frac{3}{\sqrt{5}}. \tag{9.26}$$

Variation gives the coframe equation and the connection equation. In a spinless vacuum the latter gives vanishing torsion; with spinorial matter it gives the usual Einstein-Cartan algebraic torsion source rather than identically zero torsion (Hehl et al., 1976). This qualification is necessary when the matter branch is retained.

Under the additional complete-closure reading, one retained factor is assigned to each of the 240 roots:

$$\lambda_{\text{global}} = \frac{4\pi}{\sqrt{3}} r^{240}. \tag{9.27}$$

An explicit 240-step Hamiltonian cycle in the supplied certificate proves existence of a once-per-root ordering. The association of equation (9.27) with the observed cosmological constant is interpretation-dependent; uniqueness of the cycle is unnecessary for the exponent but relevant if ordering-dependent phases are later introduced.

9.10. Stage 10: quantum gravity and complete closure

The carrier supplies three presently viable quantum readings:

1. *Covariant history reading:* each admissible carrier history h has amplitude $\exp(iS[h]/\hbar)$ and the continuum limit is a projected path integral;
2. *Operator reading:* the fixed algebra $\mathscr{B}$ carries unitary evolution while the positive decay generator grades massive off-diagonal modes;
3. *Discrete traversal reading:* root histories are fundamental and spacetime fields are collective observables of their large-scale state.

They agree on the necessary low-energy tests: a Lorentzian geometric sector, the Einstein-Cartan limit, gauge covariance, unitarity on closed systems, anomaly freedom and a controlled Standard Model limit. They differ on microscopic state space, ultraviolet observables and the meaning of the decay map. The theory-of-everything node is closed only when at least one of these readings supplies all those limits without incompatible duplicate meanings for the same carrier operation.

Framework realization theorem, present form

Grant the carrier premises of equation (9.5), the system-preserving reduction equation (9.7), the triangular-history and single-operator selection criteria, and a physical state rule whose low-energy limit is local quantum field theory. Then the chain

$$\begin{aligned} E_8 &\to A_3 \text{ decay} \to D_5 \cong SO(10) \\ &\to S(U(3)\times U(2)) \\ &\to SU(3)\times SU(2)\times U(1) \\ &\to SU(3)\times U(1)_{\mathrm{EM}}. \end{aligned}$$

is conditionally derived, one Standard Model generation with a right-handed neutrino is obtained from $\Lambda^{\mathrm{even}}\mathbb{C}^5$, and the geometry branch is generated from the same projected carrier connection. The remaining work is selection of the microscopic state rule and calculation of its spectrum, parameters and observation map.

CHAPTER 10

Focused attacks on the remaining threads

This chapter treats every surviving obligation as a mathematical object. An "attack" means: isolate the premise, compute everything that follows from it, identify the smallest unresolved map, and state a discriminator. It does not mean assigning a premature verdict.

10.1. The common normalization: coupling, Rydberg and Newton scale

After the Lie algebra and representations are fixed, a gauge action still requires an invariant bilinear form:

$$S_g = -\frac{1}{4g_U^2}\int \kappa(\mathscr{F}_{\mathrm{int}}, \star\mathscr{F}_{\mathrm{int}}). \tag{10.1}$$

Restriction of κ to the three Standard Model factors gives

$$\frac{1}{g_i^2} = \frac{k_i}{g_U^2} + \Delta_i, \qquad (k_1, k_2, k_3) = \left(\frac{5}{3}, 1, 1\right) \tag{10.2}$$

in the canonical $SU(5)$ normalization, with Δ_i collecting thresholds and kinetic mixing. The embedding fixes the ratios k_i; it does not by itself fix the common number g_U. The same normalization problem reappears in the Einstein term and in the absolute Rydberg scale. This confirms the corpus insight that the three apparent debts are one normalization map.

Three realizations remain viable:

1. *Invariant-form realization:* g_U is fixed by the normalization of the carrier trace relative to the quantum of action;
2. *Geometric-volume realization:* g_i^{-2} is a cycle or mode volume, with equation (10.2) arising from the embedding indices;
3. *Channel-sum realization:* the legacy expression

$$\alpha_{\mathrm{em}}^{-1} = \varphi^{10} + \varphi^5 + \varphi^2 + \varphi^{-2} \tag{10.3}$$

as recorded in the legacy gauge-boundary paper (Shields, 2026e). is interpreted as a finite spectral trace.

The third is presently an exact numerical ansatz, not yet a spectral derivation: the exponent set must be generated by an operator or a pre-data counting rule. A decisive calculation is therefore to construct a carrier operator K for which

$$\mathrm{Tr}_{\mathscr{H}_{\mathrm{EM}}} f(K) = \varphi^{10} + \varphi^{5} + \varphi^{2} + \varphi^{-2} \tag{10.4}$$

with the electromagnetic subspace and function f fixed independently of the measured value.

10.2. High-scale and low-scale weak angles

The framework contains two formulas that live at different scales:

$$\sin^2\theta_W(M_U) = \frac{3}{8}, \qquad \sin^2\theta_W(M_Z) \overset{\mathscr{I}}{=} \frac{W_{\mathrm{M}}}{W_{\mathrm{B}}} = \frac{1}{2\sqrt{5}}. \tag{10.5}$$

They are compatible only through a renormalization-group and threshold map. With $g_1^2 = (5/3)g'^2$,

$$\sin^2\theta_W(\mu) = \frac{3\alpha_1(\mu)}{5\alpha_2(\mu) + 3\alpha_1(\mu)}, \tag{10.6}$$

and at one loop

$$\alpha_i^{-1}(\mu) = \alpha_U^{-1} - \frac{b_i}{2\pi}\ln\frac{\mu}{M_U} - \Delta_i. \tag{10.7}$$

Consequently equation (10.5) is not a second independent derivation until one spectrum and its b_i, Δ_i connect both endpoints. The selecting calculation is a two-loop, threshold-matched run using only states already produced by the carrier decomposition. A spectrum chosen after inspecting the endpoint would be a reconstruction; a spectrum fixed by the decay operator would be a prediction.

10.3. The family spectrum and Yukawa operators

The supplied finite certificate shows that all 128 spinor-sector roots have the same leading projection norm, 3/4 (Shields, 2026c). This result has a clear interpretation: the minimal S_4-averaged reduction has an exact family symmetry, so its leading mass operator is proportional to the identity on the non-invariant family triplet,

$$M_0^2 = m_0^2 I_3. \tag{10.8}$$

Observed mass hierarchies therefore probe the next piece of structure rather than negating equation (10.8). Write

$$M_f = M_{0,f} + \epsilon_1 A_f + \epsilon_2 B_f + \cdots, \tag{10.9}$$

where every A_f, B_f must be generated by the same microscopic rule and must respect the unbroken gauge algebra. Allowing arbitrary 3×3 matrices would reintroduce the Standard Model's Yukawa freedom and leave the spectrum unconstrained.

Three retained realizations define the selector calculation:

1. *Spectral splitting:* use the general positive decay generator, rather than the minimal idempotent representative, and calculate its restriction to the family triplet;
2. *Overlap splitting:* treat Yukawa elements as triple overlaps of normalized carrier/compactification modes;
3. *History splitting:* define Y_{ij} as a sum over oriented retained histories joining left, right and order-parameter sectors.

All must return a gauge-invariant operator

$$\mathscr{L}_Y = -\overline{Q}_L Y_d H d_R - \overline{Q}_L Y_u \widetilde{H} u_R - \overline{L}_L Y_e H e_R + \text{h.c.} \tag{10.10}$$

and must predict, from pre-specified data, its singular values and relative diagonalizing matrices. This is the exact bridge to the CKM and PMNS matrices.

The legacy formulas for powers of φ, including the charged-lepton, quark, neutrino and CKM patterns, are retained as candidate spectral fingerprints (Shields, 2026h,g,b). Their next scientific upgrade is not another numerical match; it is a theorem selecting each exponent and correction sign from the spectrum of one operator. Until that selector is supplied, the formulas are interpretation-dependent reconstructions.

10.4. Chirality: from oriented cycles to an index

The finite root computation establishes that negation exchanges

$$(\mathbf{4}, \mathbf{16}) \longleftrightarrow (\overline{\mathbf{4}}, \overline{\mathbf{16}}), \tag{10.11}$$

and that a cycle followed by its reverse has zero net displacement. The corpus reads the two terms as one oriented matter history and its reverse (Shields, 2026a). To make this a field-theoretic derivation, the arrow must induce a chiral projector or a nonzero index. A sufficient target is

$$\operatorname{ind} D = \dim\ker D_L - \dim\ker D_R = 3\ \dim(\mathbf{16}) \tag{10.12}$$

after family multiplicity is included, with the conjugate modes interpreted as antiparticle states rather than independent mirror fermions.

Three mechanisms remain distinguishable:

1. an arrow-induced spectral boundary condition on the carrier Dirac operator;
2. a topological index of a compact or discrete internal complex;
3. a grading of oriented histories in which reverse cycles belong to the dual Hilbert space.

They predict different heavy mirror spectra and different anomaly-inflow structures. The index calculation and the resulting quadratic fermion action are the selectors.

10.5. The strong vacuum angle

The reversible colour-block argument has a precise consequence, but it must be stated at the correct strength. Under orientation reversal,

$$\nu[A] = \frac{1}{8\pi^2}\int \mathrm{tr}(F \wedge F) \longmapsto -\nu[A], \qquad Z(\theta) \longmapsto Z(-\theta). \tag{10.13}$$

If reversibility is an exact, unbroken symmetry, invariance implies

$$\theta \equiv -\theta \pmod{2\pi}, \qquad \text{so} \qquad \theta \in \{0, \pi\}. \tag{10.14}$$

Thus reversibility reduces a continuous parameter to two discrete possibilities. Selecting $\theta = 0$ requires one further input: positivity at the symmetric vacuum, absence of a sign-alternating topological weight, or a carrier rule that retains only the trivial orientation character. The corresponding calculation is the phase of the fermion determinant after the chiral projection. This reformulation turns the qualitative argument into a sharply bounded two-branch problem.

10.6. Complete carrier closure: existence, equivalence and phase

The certificate supplies a Hamiltonian cycle on the 240-root adjacency graph, with each step a root and the walk closing. Hence a once-per-root traversal exists and every order-independent retention product equals r^{240}.

Uniqueness has three possible meanings:

1. *Weight uniqueness:* all orderings have the same scalar weight r^{240}; already derived;

2. *Orbit uniqueness:* all admissible cycles are equivalent under the Weyl group, reversal and cyclic shift; open as a finite graph-enumeration problem;
3. *Amplitude uniqueness:* all cycles have the same phase; open unless the action is order-independent.

Only the first is needed for the scalar residue. A quantum partition function needs the third or else must sum the inequivalent phases,

$$Z_{240} = \sum_{[h]} N_{[h]} r^{240} \exp\left(\frac{iS[h]}{\hbar}\right). \tag{10.15}$$

This identifies exactly when cycle counting becomes physically relevant.

10.7. From the carrier to a controlled quantum-gravity limit

A candidate microscopic state rule must pass four nested tests:

1. *Algebraic:* gauge invariance, positivity/unitarity and anomaly cancellation;
2. *Continuum:* a locality or controlled quasi-locality theorem and a Lorentzian propagator;
3. *Semiclassical:* a state family for which

$$\langle \widehat{G}_{\mu\nu} \rangle + \Lambda g_{\mu\nu} = 8\pi G \langle \widehat{T}_{\mu\nu} \rangle + O(\hbar) \tag{10.16}$$

   holds;
4. *Infrared:* the Standard Model and general relativity with measured precision limits.

The carrier/history, operator and covariant-path-integral readings remain the three retained interpretations. A spectral dimension, two-point function and effective action calculated from the same microscopic ensemble would discriminate among them.

10.8. Cosmology: background, perturbations and observation

The geometry branch supplies Friedmann equations once homogeneity and isotropy are imposed. The proposed global residue equation (9.27) supplies a candidate vacuum parameter. A cosmological theory additionally requires:

$$\{\text{background}\} \longrightarrow \{P_{\mathscr{R}}(k), P_T(k)\} \longrightarrow \{C_\ell, P_{\mathrm{m}}(k), D_L(z)\} \longrightarrow \mathscr{L}(\text{data} \mid \vartheta). \tag{10.17}$$

The first arrow needs a perturbation state and transfer law; the second needs recombination, reaction networks and structure growth; the third needs calibrated selection functions and covariance matrices. Agreement of a background number alone does not supply these arrows.

The corpus's "pipeline closure count" is retained as an operational interpretation. It becomes testable only after a closure is defined on raw data operations before examining the inferred parameter difference. The clean protocol is: publish the counting functional, blind the cosmological result, compute counts on independently documented pipelines, then unblind and test the predeclared scaling. This separates an observer-map effect from an evolving cosmological field.

10.9. Internal self-description

The framework began with self-description; closure therefore requires an internal mathematical model of it, not merely a philosophical gloss. Let

$$X = \text{physical states}, \qquad M = \text{records}, \qquad E : X \to M, \qquad D : M \to X, \tag{10.18}$$

where E encodes and D realizes the content of a record. Let evolution with an internal record be

$$U : X \times M \to X. \tag{10.19}$$

A self-description at resolution ε satisfies

$$d_X(D(E(x)), x) \leq \varepsilon. \tag{10.20}$$

Persistent description additionally needs an injective retention map $j_t : M_t \hookrightarrow M_{t+1}$ such that

$$j_t(E_t(x_t)) \preceq E_{t+1}(U(x_t, E_t(x_t))), \tag{10.21}$$

where $\preceq$ means "is recoverable from." This states the no-overwrite condition without requiring literal storage of every microscopic bit.

Finally, physical reduction must commute with self-description. With $P : X \to X_{\text{phys}}$ and $P_M : M \to M_{\text{phys}}$, closure requires

$$PU(x, m) = U_{\text{phys}}(Px, P_M m), \qquad P_M E(x) = E_{\text{phys}}(Px). \tag{10.22}$$

Equation (10.22) is the internal-self-description version of the commuting realization criterion: the world described after reduction must equal the reduction of the world and its internal record evolved together.

Three interpretations are retained:

1. *Fixed-point semantics:* a self-description is a stable or approximately stable point of $D \circ E$;

2. *Coalgebraic process:* E exposes the observable continuation of a state, and behavioural equivalence replaces literal identity;
3. *Quantum instrument:* E is a completely positive instrument whose classical outcome record and post-measurement state jointly satisfy equations (10.21) and (10.22).

These readings coincide at the abstract commuting-square level. They differ on record capacity, reversibility and disturbance, which makes those quantities the correct discriminants.

10.10. Consolidated frontier

TABLE 10.1. What has been learned by attacking each remaining thread.

Thread	Narrowed status	Selecting calculation
Common coupling	one normalization, three realizations	derive carrier trace/volume/spectral sum before comparison
Weak angle	high- and low-scale formulas can be endpoints of one flow	two-loop running with carrier-produced spectrum and thresholds
Mass hierarchy	leading S_4 average gives an exact degenerate triplet	spectrum of the general decay generator or a derived overlap/history operator
CKM/PMNS	relative diagonalization problem isolated	compute all Yukawa matrices from the same pre-data rule
Chirality	reversal fact certified; index bridge isolated	construct Dirac operator and evaluate its index and quadratic action
Strong angle	continuous ambiguity reduced to $\{0, \pi\}$	determinant phase or orientation-character calculation
Carrier closure	scalar weight unique; orbit and phase questions separated	enumerate cycle orbits only if the action depends on order

Thread	Narrowed status	Selecting calculation
Quantum gravity	three microscopic readings share explicit low-energy tests	spectral dimension, correlator and effective-action calculation
Cosmology	background and observation chain separated	derive perturbations and preregister the operational pipeline map
Self-description	encoder, decoder, retention and commuting square specified	construct one nontrivial model satisfying all four simultaneously

No item in table 10.1 is an unstructured gap. Each is now a finite calculation, a spectral problem, a continuum-limit theorem, or a preregistered empirical comparison. That is the strongest scientifically useful form an open interpretation can presently take.

APPENDIX A

Complete derivation and interpretation ledger

A.1. Image-node coverage

TABLE A.1. Coverage of every labelled node in the supplied image.

ID	Node	Present status	Next selecting calculation
N01	Theory of everything	common closure graph and commuting-realization criteria enacted	complete the localized mass, constitutive, kernel, and prospective-measurement frontier
N02	Quantum gravity	three realization classes retained with common tests	construct correlators, continuum limit and effective action
N03	Spacetime curvature	weighted-square route varied to the registered Einstein–Cartan equations	fix normalization and test controlled gravitational limits
N04	GUT sector	$E_8 \to A_3$ decay $\to D_5 \cong SO(10)$, chiral 16, and anomaly chain fixed	threshold spectrum, normalization, and microscopic index realization

ID	Node	Present status	Next selecting calculation
N05	Standard cosmology	FLRW background, dark-sector perturbations, recombination, and CAMB transfer computed	public likelihood, independent recombination cross-run, and microscopic matter identity
N06	Standard Model	gauge algebra, chiral matter, charges, anomalies, and one-loop coefficients fixed	stationary mass spectrum, thresholds, and mixing matrices
N07	Strong $SU(3)$	Yang–Mills action, representations and running derived	nonperturbative observables and normalization selector
N08	Electroweak	symmetry reduction complete; three order-parameter readings retained	select the order parameter and compute thresholds
N09	Weak	charged/neutral fields and Fermi limit derived	chiral index, flavour and radiative corrections
N10	Electromagnetism	unbroken generator and Maxwell action derived	common coupling normalization and constitutive test
N11	Electricity	derived as observer split of F	none at the kinematic level
N12	Magnetism	derived as observer split of F	none at the kinematic level

A.2. Relation coverage

TABLE A.2. Coverage of every visible relation and required cross-edge.

ID	Relation	Type	Status
E01	TOE–quantum gravity	closure	dependency and internal-self-description conditions explicit
E02	quantum gravity–curvature	correspondence limit	semiclassical target specified; microscopic derivation open
E03	quantum gravity–GUT	common quantum dynamics	carrier action common; three microscopic readings retained
E04	curvature–cosmology	symmetry reduction	FLRW/Friedmann background derived
E05	GUT–Standard Model	subgroup branching	exact A_3-decay, D_5, $3+2$, charge and anomaly route displayed
E06	Standard Model–strong/electroweak	gauge-sector decomposition	complete at Lie-algebra and action level
E07	electroweak–weak/electromagnetic	vacuum reduction	neutral mass matrix diagonalized
E08	electromagnetism–electricity/magnetism	observer decomposition	covariant split and reconstruction derived
X01	matter–curvature	variational coupling	stress–energy and conservation derived
X02	quantum fields–curved spacetime	semiclassical bridge	requirement equation specified
X03	particle physics–cosmology	thermal history	non-equilibrium recombination and full linear transfer computed; precision likelihood active
X04	theory carrier–physical sectors	realization map	conditional expectation, commutant and off-diagonal maps derived; boundary mass operator located; stationary selector active

A.3. Rule for future entries

No row may be closed by assertion. Closure requires a displayed derivation or a reproducible certificate, an explicit assumption list, and a check that all competing retained interpretations either agree on the result or are separated by a stated observable. A non-unique interpretation remains open under its declared premises.

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